

Solver Documentation
electroChem
carbonEq branch

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1 Introduction

This document documents the custom OpenFOAM solver electroChem made for simulating reacting multicomponent species electrolytic systems with the goal of supporting microbial electrosynthesis of CO_2 to acetate. The solver is an extension of the existing pimpleFOAM FVM solver from OpenFOAM.

1.1 carbonEq branch

This branch adds an equilibrium projection model for the combined water and inorganic carbon equilibrium. This model aims to resolve the equilibrium species distribution in a way that satisfies mass conservation though reaction extend.

2 Governings equations

The flow is treated as incompressible and solved through the incompressible Navier Stokes formulation using the PIMPLE algorithm

$$\nabla \cdot \mathbf{u} = 0 \quad (1)$$

$$\frac{\partial \mathbf{u}}{\partial t} + \nabla \cdot (\mathbf{u}\mathbf{u}) = -\frac{1}{\rho} \nabla P + \nu \nabla^2 \mathbf{u} + \mathbf{g} \quad (2)$$

All active species concentrations are treated by reactive Nerst-Planck transport

$$\frac{\partial c_i^*}{\partial t} + \nabla \cdot (c_i^* \mathbf{u}) = \nabla \cdot (D_i \nabla c_i^*) + \nabla \cdot \left(D_i \frac{z_i F}{RT} c_i^* \nabla \phi \right) + R_i \quad (3)$$

Where c_i^* is the transported species concentration, R_i is the reactive species source term combining the contributions of all kinetic reactions

$$R_i = \sum_j \nu_{ij} \dot{r}_j \quad (4)$$

where ν_{ij} is the stoichiometric constant of species "i" for reaction "j". After transport fast equilibrium reactions are assumed to instantaneously settle giving projected equilibrium concentration c_i^{eq} from which the potential is update. If no equilibrium reactions are included, then $c_i^{eq} = c_i^*$. A constant current density is prescribed over the domain in the electrode normal direction $\mathbf{e}_j$

$$\mathbf{j} = j_{app} \cdot \mathbf{e}_j \quad (5)$$

With this the potential gradient can be found directly from the current equation

$$\mathbf{j} = -F \sum_i z_i D_i \nabla c_i^{eq} - \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{eq} \nabla \phi \quad (6)$$

defining

$$\kappa = \frac{F^2}{RT} \sum_i z_i^2 D_i c_i^{eq} \quad (7)$$

and

$$\mathbf{j}_D = -F \sum_i z_i D_i \nabla c_i^{eq} \quad (8)$$

and rewriting equation (6) as

$$\nabla \phi = \frac{\mathbf{j}_D - \mathbf{j}}{\kappa} = \frac{\mathbf{j}_D - j_{app} \cdot \mathbf{e}_j}{\kappa} \quad (9)$$

Species are converted to mass fraction

$$Y_i = c_i^{eq} \frac{M_i}{\rho} \quad (10)$$

and the inert species "I" is explicitly found by

$$Y_I = 1 - \sum_{i \neq I} Y_i \quad (11)$$

pH is the log concentration of H^+ in mol/L

$$pH = -\log_{10} \left(\frac{c_{H^+}}{1000} \right) \quad (12)$$

3 Custom boundary conditions

This section covers the custom boundary condition and helper functions added with this model.

3.1 Electrode boundaries

These boundary condition determines the boundary flux at the electrode patches.

3.1.1 MolarFlux

The electrode flux is comprised of a faradaic source combined with diffusive and migration flux assuming no penetration of the electrode.

$$\nabla c_i \cdot \mathbf{n} = \frac{\nu_i j_{el}}{D_i n F} - \left(\frac{z_i F}{RT} c_i \nabla \phi \right) \quad (13)$$

Discretization

The boundary is solved using a Robin equation formulation. Introduce

$$A = \frac{\nu_i j}{n F D_i}, \quad \beta = \frac{z_i F}{RT} (\nabla \phi \cdot \mathbf{n}) \quad (14)$$

At the electrode boundary

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (15)$$

where $c_{b,i}$ is the boundary concentration of species i . Then the boundary adjacent gradient is defined as

$$\nabla c_i = \delta (c_{b,i} - c_{p,i}) \quad (16)$$

where $\delta = 1/\Delta_n$, Δ_n is the distance from boundary to cell-center and $c_{p,i}$ is the boundary adjacent cell-center concentration. Insert into equation 15

$$\delta (c_{b,i} - c_{p,i}) = A - \beta c_{p,i} \quad (17)$$

isolate boundary concentration

$$c_{b,i} = \frac{\delta c_{p,i} + A}{\delta + \beta} \quad (18)$$

then the boundary flux becomes

$$\nabla c_i \cdot \mathbf{n} = A - \beta c_{b,i} \quad (19)$$

3.2 Inlet boundaries

These custom boundaries are treats the inlet patch and simplifies some of the commonly used boundary conditions.

3.2.1 pHMolarConcentration

This custom boundary automatically determines the distribution of H^+ and OH^- ions at a given pH. c_{H^+} is directly treated by

$$c_{H^+} = 10^{-pH} \quad (20)$$

and c_{OH^-} by

$$c_{OH^-} = \frac{K_W}{c_{H^+}} \quad (21)$$

The user points the BC to the relevant equilibrium model from which K_W is read to ensure internal consistency.

3.2.2 inertMolarConcentration

This custom boundary condition is used for the inert species and sets the inlet concentration as the remainder such that the total mass-fraction of the system adds up to 1. The remainder for the inert species "I" is defined

$$Y_I = 1 - \sum_{i \neq I} Y_i \quad (22)$$

Concentration fields are converted to mass-fraction using

$$Y_i = \frac{c_i M_i}{\rho} \quad (23)$$

followed by

$$c_I = \frac{\rho Y_I}{M_I} \quad (24)$$

So directly

$$c_I = \frac{\rho}{M_I} \left(1 - \sum_{i \neq I} \frac{c_i M_i}{\rho} \right) \quad (25)$$

4 Reaction Models

The solver supports kinetic reactions treated as source terms or instantaneous equilibrium projections applied after species transport.

4.1 Kinetic Reactions

Kinetic reactions are for reactions developing on timescales comparable or slower than that of the transport and must be treated explicitly.

4.1.1 Reversible kinetic reaction

The reaction rate of reversible kinetic reactions are found as

$$\dot{r} = k_f \prod c_R - k_b \prod c_P \quad (26)$$

where k_f is the forward reaction rate and k_b is the reverse reaction rate. c_R is the concentration of reactants and c_P is the concentration of products. This model can allow for negative concentrations when used for fast equilibrium models.

4.1.2 Multiplicative Monod reaction

Microbial reactions are described through multiplicative Monod

$$\dot{r} = X \dot{q}_{max} \left(\frac{c_d}{c_d + K_{S,d}} \right) \left(\frac{c_a}{c_a + K_{S,a}} \right) \quad (27)$$

where X is the biomass density, $\dot{q}_{max}$ is the maximum specific consumption rate, K_S is the half saturation constant and the subscripts d and a describe the electron donor and acceptor substrates.

4.2 Equilibrium Reactions

Equilibrium reactions assume that the equilibrium settles much faster than the transport time scale and can therefore be assumed instantaneously before the next transport step.

4.2.1 Water equilibrium

The water equilibrium model projects the local water equilibrium under the constraint that the local ion difference is preserved as the reaction can only happen for ion pairs. The equilibrium reaction of water is



with equilibrium constant

$$K_W = c_{H^+} c_{OH^-} \quad (28)$$

Assuming water is treated as an inert species and does not actively participate. The ion difference is defined as

$$A = c_{H^+} - c_{OH^-} \quad (29)$$

Rewriting and substituting equation 29 into equation 34 the following algebraic concentration projections are found

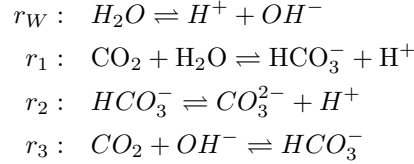
$$c_{H^+} = \frac{A + \sqrt{A^2 + 4K_w}}{2} \quad (30)$$

$$c_{OH^-} = \frac{-A + \sqrt{A^2 + 4K_w}}{2} \quad (31)$$

This model can allow for negative concentrations for compensating for overshooting slow consumptions, such as electrode consumption.

4.2.2 Carbon water equilibrium

This model treats the combined water and carbonate species equilibria. The model splits the reactions into kinetically resolved slow reactions and fast equilibrium projections. The equilibrium projection is solved by the reactions extents through the Newton method. The full reaction scheme is



The slow reactions r_1 and r_3 are treated as explicit kinetic reactions

$$\dot{r}_1 = k_{f,1} c_{CO_2} - \frac{k_{b,1}}{K_{1,2}} c_{HCO_3^-} c_{H^+} \quad (32)$$

$$\dot{r}_3 = k_{f,3} c_{CO_2} c_{OH^-} - k_{b,3} c_{HCO_3^-} \quad (33)$$

See appendix A.1.1 to see simplifications to reaction system. This leaves two fast equilibrium reactions r_W and r_2 and four equilibrium variables c_{H^+} , c_{OH^-} , $c_{HCO_3^-}$, $c_{CO_3^{2-}}$. These are treated through reaction extent to satisfy the equilibrium condition of both reactions

$$K_W = c_{H^+} c_{OH^-} \quad (34)$$

$$K_2 = \frac{c_{CO_3^{2-}} c_{H^+}}{c_{HCO_3^-}} \quad (35)$$

For each equilibrium species this starts from the transported concentration c_i^* and tells how far the fast reaction must proceed to satisfy the equilibrium. From the equilibrium reaction stoichiometry, the reaction extents are constrained by

$$c_{H^+}^{eq} = c_{H^+}^* + \xi_W + \xi_2 \quad (36)$$

$$c_{OH^-}^{eq} = c_{OH^-}^* + \xi_W \quad (37)$$

$$c_{HCO_3^-}^{eq} = c_{HCO_3^-}^* - \xi_2 \quad (38)$$

$$c_{CO_3^{2-}}^{eq} = c_{CO_3^{2-}}^* + \xi_2 \quad (39)$$

This system can be rewritten into a non-linear equation system of the reaction extends. By rewriting 34 and substituting 36 and 37 an expression for the water equilibrium by reaction extend is found

$$F_W(\xi_W, \xi_2) = (c_{H^+}^* + \xi_W + \xi_2)(c_{OH^-}^* + \xi_W) - K_W = 0 \quad (40)$$

and for bicarbonate dissociation by rewriting 35 and substituting 36, 38, 39

$$F_2(\xi_W, \xi_2) = (c_{CO_3^{2-}}^* + \xi_2)(c_{H^+}^* + \xi_W + \xi_2) - K_2(c_{HCO_3^-}^* - \xi_2) = 0 \quad (41)$$

This is now defined as a non-linear equation system

$$\xi = \begin{bmatrix} \xi_W \\ \xi_2 \end{bmatrix}, \quad \mathbf{F}(\xi) = \begin{bmatrix} F_W(\xi_W, \xi_2) \\ F_2(\xi_W, \xi_2) \end{bmatrix} \quad (42)$$

And the goal is to find ξ such that the following is satisfied

$$\mathbf{F}(\xi) = 0$$

To find the value of ξ that satisfies $\mathbf{F}(\xi) = 0$ the two-variable newton method is used. Here at at given iteration "n"

$$\xi^n = \begin{bmatrix} \xi_W^n \\ \xi_2^n \end{bmatrix}$$

where $\mathbf{F}(\xi^n) \neq 0$, the goal is to estimate how much ξ should change to get closer to $\mathbf{F}(\xi) = 0$

$$\xi^{n+1} = \xi^n + \Delta\xi \quad (43)$$

This change is defined as the newton increment

$$\Delta\xi = \begin{bmatrix} \Delta\xi_W \\ \Delta\xi_2 \end{bmatrix}$$

The newton increment is approximated through

$$\mathbf{J}\Delta\xi = -\mathbf{F} \quad (44)$$

where "J" is the Jacobian. This expands to the equations system

$$J_{11}\Delta\xi_W + J_{12}\Delta\xi_2 = -F_W \quad (45)$$

$$J_{21}\Delta\xi_W + J_{22}\Delta\xi_2 = -F_2 \quad (46)$$

The solution of which is

$$\Delta\xi_W = \frac{-F_W J_{22} + J_{12} F_2}{D} \quad (47)$$

$$\Delta\xi_2 = \frac{J_{21} F_W - J_{11} F_2}{D} \quad (48)$$

the derivation can be found in appendix A.1.2. The Jacobian for this system is

$$\mathbf{J} = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} = \begin{bmatrix} c_{H^+} + c_{OH^-} & c_{OH^-} \\ c_{CO_3^{2-}} & c_{CO_3^{2-}} + c_{H^+} + K_2 \end{bmatrix} \quad (49)$$

see appendix A.1.3. with determinant

$$D = J_{11}J_{22} - J_{12}J_{21}$$

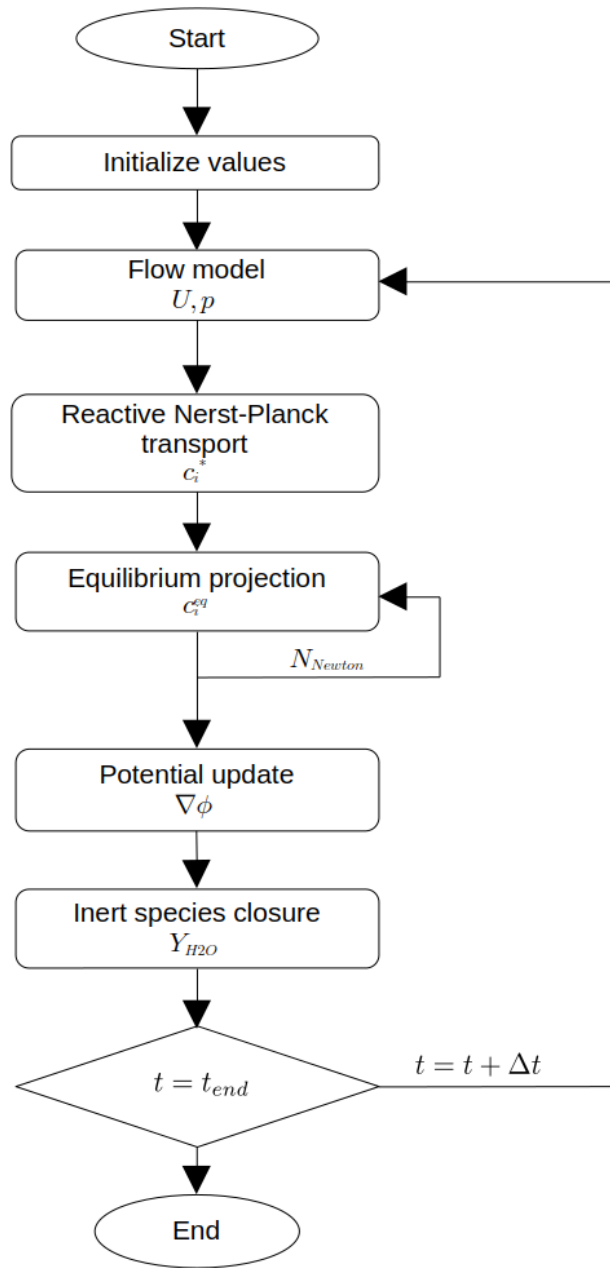
From this the reaction extends are iterated according to equation 43. This is iterated until the reaction extends ξ is converged. When converged the concentrations are updated using

$$\mathbf{c}^{eq} = \mathbf{c}^* + N\xi \quad (50)$$

Where N is the stoichiometric coefficients of the equilibrium reactions

$$N = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & -1 \\ 0 & 1 \end{bmatrix} \quad (51)$$

5 Algorithm flowchart



6 Solver structure

Source code structure

```
electroChem/  
|-- Documentation/  
|-- Make/  
|   |-- files  
|   |-- options  
|-- reactionModels/  
|   |-- equilibriumReactions/  
|       |-- waterEquilibrium.C  
|       |-- waterEquilibrium.H  
|       |-- waterCarbonEquilibrium.C  
|       |-- waterCarbonEquilibrium.H  
|   |-- kineticReactions/  
|       |-- multiplicativeMonod.C  
|       |-- multiplicativeMonod.H  
|       |-- reversibleKineticReaction.C  
|       |-- reversibleKineticReaction.H  
|-- cEqn.H  
|-- createFields.H  
|-- diagnostics.H  
|-- electroChem.C  
|-- pEqn.H  
|-- phiEqn.H  
|-- UEqn.H  
|-- YEqn.H
```

custom functions

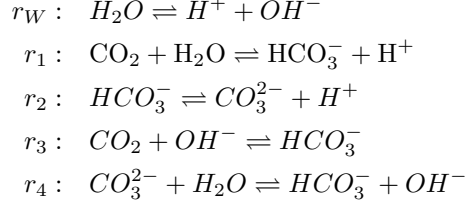
```
src/  
|-- customLibs/  
|   |-- electrochemistry/  
|       |-- derivedFvPatchFields/  
|           |-- MolarFlux/  
|               |-- faradaicMolarFlux.C  
|               |-- faradaicMolarFlux.H  
|               |-- robinMolarFlux.C  
|               |-- robinMolarFlux.H  
|           |-- MolarConcentration/  
|               |-- pHMolarConcentration.C  
|               |-- pHMolarConcentration.H  
|               |-- inertMolarConcentration.C  
|               |-- inertMolarConcentration.H  
|-- Make/  
|   |-- files  
|   |-- options
```

A Appendix

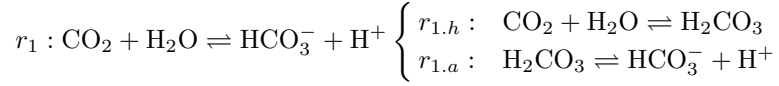
A.1 Carbon water equilibrium

A.1.1 Reaction simplifications

Full reaction scheme



The CO_2 -bicarbonate equilibrium r_1 consists of a lumped hydration step $r_{1,h}$ and acid dissociation step $r_{1,a}$



The CO_2 hydration step as well as the hydroxide path r_3 are both slow reactions and cannot comfortably be treated as instantaneous equilibrium projections and must be treated kinetically

$$\dot{r}_{1,h} = k_{f,1}c_{CO_2} - k_{b,1}c_{H_2CO_3} \quad (52)$$

$$\dot{r}_3 = k_{f,3}c_{CO_2}c_{OH^-} - k_{b,3}c_{HCO_3^-} \quad (53)$$

reactions $r_{1,h}$ and $r_{1,a}$ can be thermodynamically lumped into a single kinetic equation.

$$K_{1,2} = \frac{c_{HCO_3^-}c_{H^+}}{c_{H_2CO_3}} \Rightarrow c_{H_2CO_3} = \frac{c_{HCO_3^-}c_{H^+}}{K_{1,2}} \quad (54)$$

$$\dot{r}_1 = k_{f,1}c_{CO_2} - \frac{k_{b,1}}{K_{1,2}}c_{HCO_3^-}c_{H^+} \quad (55)$$

Effectively eliminating H_2CO_3 as a variable and making CO_2 treated entirely kinetically. Then the renaming equilibrium reactions are

$$K_W = c_{H^+}c_{OH^-} \quad (56)$$

$$K_2 = \frac{c_{CO_3^{2-}}c_{H^+}}{c_{HCO_3^-}} \quad (57)$$

$$K_4 = \frac{c_{HCO_3^-}c_{OH^-}}{c_{CO_3^{2-}}} \quad (58)$$

Where r_4 can be shown to be redundant. The equilibrium equations for water and bicarbonate-carbonate are rewritten

$$K_2 = \frac{c_{CO_3^{2-}}c_{H^+}}{c_{HCO_3^-}} \Rightarrow \frac{c_{HCO_3^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}}{K_2} \quad (59)$$

$$K_W = c_{H^+}c_{OH^-} \Rightarrow c_{OH^-} = \frac{K_W}{c_{H^+}} \quad (60)$$

The rewritten forms are multiplied and H^+ cancel out on the right-hand side

$$\frac{c_{HCO_3^-}c_{OH^-}}{c_{CO_3^{2-}}} = \frac{c_{H^+}K_W}{K_2c_{H^+}} = \frac{K_W}{K_2} \quad (61)$$

Showing the left-hand side is exactly the equilibrium constant for r_4

$$K_4 = \frac{K_W}{K_2} \quad (62)$$

making it a linear combination of the other equilibrium reactions and thereby redundant as they are not resolved kinetically but projected.

A.1.2 Equilibrium projection

Given the non-linear equation system

$$F_W(\xi_W, \xi_2) = (c_{H^+}^* + \xi_W + \xi_2)(c_{OH^-}^* + \xi_W) - K_W = 0 \quad (63)$$

$$F_2(\xi_W, \xi_2) = (c_{CO_3^{2-}}^* + \xi_2)(c_{H^+}^* + \xi_W + \xi_2) - K_2(c_{HCO_3^-}^* - \xi_2) = 0 \quad (64)$$

$$\xi = \begin{bmatrix} \xi_W \\ \xi_2 \end{bmatrix}, \quad \mathbf{F}(\xi) = \begin{bmatrix} F_W(\xi_W, \xi_2) \\ F_2(\xi_W, \xi_2) \end{bmatrix} \quad (65)$$

To estimate this newton increment the non-linear function is approximated by first order Taylor-expansion where "J" is the Jacobian

$$\mathbf{F}(\xi^n + \Delta\xi) \approx \mathbf{F}(\xi^n) + \mathbf{J}(\xi^n)\Delta\xi \quad (66)$$

The goal is then to minimize this expression to approach equilibrium

$$\mathbf{F}(\xi^n + \Delta\xi) = 0 \quad (67)$$

With the Newton approximation

$$\mathbf{F}(\xi^n) + \mathbf{J}(\xi^n)\Delta\xi \approx 0 \quad (68)$$

Which is rearranged to

$$\mathbf{J}\Delta\xi = -\mathbf{F} \quad (69)$$

Which is the form that is solved. The next step is to derive the Jacobian, containing the partial derivatives of F_W and F_2 to ξ_W and ξ_2

$$\mathbf{J} = \begin{bmatrix} \frac{\partial F_W}{\partial \xi_W} & \frac{\partial F_W}{\partial \xi_2} \\ \frac{\partial F_2}{\partial \xi_W} & \frac{\partial F_2}{\partial \xi_2} \end{bmatrix} \quad (70)$$

Where

$$J_{11} = \frac{\partial F_W}{\partial \xi_W}, \quad J_{12} = \frac{\partial F_W}{\partial \xi_2} \quad (71)$$

$$J_{21} = \frac{\partial F_2}{\partial \xi_W}, \quad J_{22} = \frac{\partial F_2}{\partial \xi_2} \quad (72)$$

Then

$$\begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix} \begin{bmatrix} \Delta\xi_W \\ \Delta\xi_2 \end{bmatrix} = - \begin{bmatrix} F_W \\ F_2 \end{bmatrix} \quad (73)$$

which simplifies to

$$J_{11}\Delta\xi_W + J_{12}\Delta\xi_2 = -F_W \quad (74)$$

$$J_{21}\Delta\xi_W + J_{22}\Delta\xi_2 = -F_2 \quad (75)$$

Which is a 2x2 non-linear problem. The system expressed on general form is

$$ax + by = r \quad (76)$$

$$cx + dy = s \quad (77)$$

And determinant

$$D = ad - bc \quad (78)$$

Then the solution is

$$x = \frac{rd - bs}{D}, \quad y = \frac{as - rc}{D} \quad (79)$$

Inserting the Newton system values

$$a = J_{11}, \quad b = J_{12}, \quad c = J_{21}, \quad d = J_{22}, \quad r = -F_W, \quad s = -F_2$$

and

$$D = J_{11}J_{22} - J_{12}J_{21}$$

Then the newton increments become

$$\Delta\xi_W = \frac{-F_W J_{22} + J_{12} F_2}{D} \quad (80)$$

$$\Delta\xi_2 = \frac{J_{21} F_W - J_{11} F_2}{D} \quad (81)$$

A.1.3 Jacobion

Function equations

$$F_W(\xi_W, \xi_2) = (c_{H^+}^* + \xi_W + \xi_2)(c_{OH^-}^* + \xi_W) - K_W$$

$$F_2(\xi_W, \xi_2) = (c_{CO_3^{2-}}^* + \xi_2)(c_{H^+}^* + \xi_W + \xi_2) - K_2(c_{HCO_3^-}^* - \xi_2) = 0$$

Partial derivatives

$$J_{11} = \frac{\partial F_W}{\partial \xi_W} = c_{H^+}^* + c_{OH^-}^* + 2\xi_W + \xi_2 = c_{H^+} + c_{OH^-} \quad (82)$$

$$J_{12} = \frac{\partial F_W}{\partial \xi_2} = c_{OH^-}^* + \xi_W = c_{OH^-} \quad (83)$$

$$J_{21} = \frac{\partial F_2}{\partial \xi_W} = c_{CO_3^{2-}}^* + \xi_2 = c_{CO_3^{2-}} \quad (84)$$

$$J_{22} = \frac{\partial F_2}{\partial \xi_2} = c_{CO_3^{2-}}^* + c_{H^+}^* + \xi_W + 2\xi_2 + K_2 = c_{CO_3^{2-}} + c_{H^+} + K_2 \quad (85)$$

Expressed on matrix form gives

$$\mathbf{J} = \begin{bmatrix} c_{H^+} + c_{OH^-} & c_{OH^-} \\ c_{CO_3^{2-}} & c_{CO_3^{2-}} + c_{H^+} + K_2 \end{bmatrix} \quad (86)$$